

GEA

Global Earthquakes Analytics

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Abstract

This paper presents GEA, a visual analytics tool that integrates multiple techniques to gain insights into the seismic activity observed throughout the year 2023. The core features include a geographical map, a parallel coordinates plot, a heatmap, and a t-distributed Stochastic Neighbor Embedding (t-SNE) scatterplot to identify patterns, trends, and potential hidden relations associated with an earthquake event. Users, like Seismologists and Geophysicists, can interact with the system for a focused analysis and explore insights from them conducting in-depth research and forecasting. The tool also presents opportunities for further enhancements and collaboration with international seismic networks trying to reduce the impact of seismic events on human life, implementing better early-warning systems in high-risk zones and safer infrastructure.

1 Introduction

The analysis of earthquake data plays a crucial role in comprehending the complex dynamics of seismic activity and in improving disaster preparedness strategies. Earthquakes, as unpredictable natural phenomena, present significant threats to human life, infrastructure, and the global economy. A comprehensive study of earthquake data allows for valuable insights into seismic patterns, enabling better risk assessment and the enhancement of early warning systems. This analysis also informs the development of resilient infrastructure and effective disaster response plans, particularly in areas prone to high seismic activity. Understanding key factors such as frequency, magnitude, and depth of earthquakes is essential for minimizing potential damage and ensuring the safety of communities. Moreover, seismic data extends its significance beyond immediate disaster response. Long-term analysis assists researchers in identifying high-risk areas, refining building codes, and formulating policies to mitigate the impact of future earthquakes. Furthermore, earthquake data enables scientists to study tectonic processes, assess the behavior of fault lines, and monitor volcanic activity. These insights contribute to a comprehensive approach to disaster management, urban planning, and public safety initiatives in seismically active regions. The Earthquakes 2023 Global Dataset, sourced from Kaggle [1], offers a comprehensive collection of global seismic activity data throughout the year 2023. This dataset includes essential details such as magnitude, depth, geographical coordinates, and the type of seismic event (e.g., earthquake or volcanic eruption). Additionally, it provides proximity to the nearest reporting stations, offering valuable

context for understanding the precision and timeliness of event detection. Covering 20000 events, the dataset serves as a rich resource for exploring global earthquake patterns and identifying trends in seismic activity across various tectonic settings. For this paper, a subset of the dataset has been thoughtfully selected and processed to ensure it aligns with the specific analytical objectives. To make this wealth of seismic data more accessible and actionable, I'm introducing GEA (Global Earthquakes Analytics), an advanced visualization tool designed to enable in-depth exploration of global seismic events. GEA offers an interactive platform where users can analyze earthquake data across multiple dimensions, such as geographic distribution, magnitude, depth, and time. By transforming raw data into meaningful visualizations, GEA provides critical insights for a wide range of users, including seismologists, urban planners, policymakers, and emergency management professionals. One of the core features of GEA is its geographical map, which displays seismic events on a global map. In addition to spatial analysis, GEA offers a parallel coordinates plot that visualizes relationships between various seismic attributes. Temporal analysis is facilitated through the heatmap feature, which presents seismic events across a timeline. For dimensionality reduction and clustering analysis, GEA also incorporates a t-SNE scatterplot, which visualizes seismic events based on their characteristics. All these visualizations are described in detail later in Section 4. Through its comprehensive and interactive design, GEA offers a powerful tool for analyzing seismic data and gaining actionable insights. Whether used by researchers to study tectonic activity, policymakers for disaster preparedness, or emergency response teams for event monitoring,

GEA facilitates a more nuanced understanding of global earthquake patterns. Its ability to integrate spatial, temporal, and multi-dimensional data into an easy-to-use platform makes it an essential resource for improving earthquake risk assessment, enhancing resilience in vulnerable regions, and ultimately contributing to safer communities worldwide.

2 Related works

Several tools and platforms have been developed to visualize and analyze earthquake data, offering users insights into global seismic activity. Among the most prominent are the USGS Earthquake Map, the IRIS Earthquake Browser, and the Worldwide Earthquakes Visualization by Tableau. Each of these tools provides unique features for visualizing seismic data and serves different audiences with varying levels of analytical depth.

2.1 USGS earthquake map

The USGS Earthquake Map [2] is one of the most widely used tools for real-time earthquake monitoring. Hosted by the United States Geological Survey (USGS), this tool allows users to visualize earthquakes across the globe in real time. It provides detailed information about each seismic event, such as magnitude, depth, location, and time of occurrence. The interface is primarily map-based, where earthquakes are represented as points, and users can filter data by date, magnitude, and region. This tool is valuable for emergency responders, geologists, and the general public who need quick access to the latest earthquake events. However, it is largely focused on real-time data, and while it offers some filtering options, its analytical capabilities are somewhat limited compared to more in-depth data exploration tools.

2.2 IRIS earthquake browser

The IRIS Earthquake Browser (IEB) [3] is a more specialized tool that allows users to browse earthquake data from various global networks, including the USGS. The browser offers extensive filtering capabilities based on event parameters such as magnitude, depth, time range, and geographical location. Users can explore seismic events in a customizable map view or through a detailed event list. The IEB also provides users with the option to download data in text or CSV format for further analysis, making it a useful resource for researchers. However, like the USGS tool, the IEB's visualizations are primarily map-based, and while it allows users to delve into historical data, it does not offer advanced multi-

dimensional analysis or interactivity beyond basic event filters.

2.3 Worldwide earthquakes visualization

The Worldwide Earthquakes Visualization [4] created using Tableau offers a visually appealing, interactive dashboard that allows users to explore global earthquake data. The tool uses a combination of map visualizations, bar charts, and scatter plots to present earthquake information in a user-friendly format. Users can filter by magnitude and depth, and the dashboard adjusts to display the filtered data. This tool is designed to make earthquake data accessible to a broad audience, including the general public and data enthusiasts. While it provides a compelling visual experience, its analytical depth is limited compared to more research-focused platforms, as it lacks more advanced multi-dimensional analysis and correlation capabilities.

2.4 Comparison with GEA

While tools like the USGS Earthquake Map, IRIS Earthquake Browser, and Worldwide Earthquakes Visualization offer useful seismic data, GEA stands out for its advanced, multi-dimensional analysis features:

1. Interactive data exploration: USGS and IRIS rely on basic map visualizations, while GEA offers deeper interaction through the geographical map, parallel coordinates, heatmap, and t-SNE scatterplot, enabling users to explore relationships between variables like magnitude, depth, and station proximity.
2. Multi-dimensional analysis: unlike the limited filtering in USGS and IRIS, GEA's parallel coordinates plot allows users to analyze complex relationships between multiple earthquake attributes, making it ideal for researchers and advanced users.
3. Temporal and spatial correlation: GEA's heatmap feature enhances temporal analysis, allowing users to identify clusters and spatial patterns, providing richer insights into seismic activity over time.
4. Dimensionality reduction: GEA's t-SNE scatterplot offers unique clustering capabilities, allowing the grouping of seismic events based on multiple characteristics for deeper analysis, which is unavailable in USGS, IRIS, or other tools.
5. Comparative insights: while other tools focus on isolated attributes like location or magnitude, GEA's integrated approach allows users to compare variables (e.g., magnitude and depth) over time, offering more comprehensive insights.

In conclusion, GEA provides a more advanced, interactive, and multi-variable analysis compared to USGS, IRIS, and Worldwide Earthquakes Visualization, offering richer insights for understanding seismic events.

3 Dataset

The Earthquakes 2023 Global dataset provides a comprehensive collection of seismic events that occurred globally in 2023. It includes detailed information on various attributes of each earthquake, such as location (latitude and longitude), depth, magnitude, related errors in measurements, reporting network, and date and time of occurrence. This dataset is ideal for analyzing global seismic trends, studying the geographical distribution of earthquakes, and understanding their frequency and intensity. It can also be used for machine learning tasks like earthquake prediction models. The dataset is designed to be user-friendly, allowing for easy integration into data science workflows and in-depth scientific analyses. Given its global coverage and detailed attributes, it is valuable for both researchers and developers in the field of seismology and natural hazard studies.

3.1 Data preprocessing

The selected dataset provided over 20,000 tuples of 22 columns. However, since the whole dataset was too big for application purposes, I have decided to consider a smaller subset of only 5000 rows. This smaller subset perfectly fits the needs of the application, avoiding crashes and the unresponsiveness of the UI. After the selection of rows, I properly polished the dataset dropping duplicates and rows with null values, by removing the columns *nst*, *gap*, *rms*, *id*, *updated*, and *status* because I considered them not useful for the tool. Additionally, I have enumerated all the rows starting from 0 to give them a unique identifier that is more readable than the event id. Then, I grouped all the time occurrences into months and weeks for a clear separation of events and categorized all the magnitude, depth, and epicenter distances into standard categories. After that, I have grouped in sets all the values of all the categorical attributes, to visualize them in a parallel coordinates chart, to let the user have a better understanding of some relation between these attributes. Finally, I have applied a dimensionality reduction technique to show some characteristics in a scatterplot chart. In particular, I have used the t-SNE dimensionality reduction technique applied to the following attributes of the data: *mag*, *depth*, *latitude*, *longitude*, *magType*, *magSource*, *type*, *depthError*, *magError*, *magNst*. I have

applied t-SNE to these attributes because they were numerical attributes, or categorical but properly encoded, to find some clusters that could help the intended user to find some insights about different magnitude categories. Additionally, I have chosen to apply t-SNE instead of PCA or MDS because, although it introduces the possibility of having false positives and false negatives, it amplifies the separation between clusters of the data, that are arranged on the 2D space of the scatterplot chart by iterating and minimizing a stress function. Since I aimed to find clusters of the data to have a better understanding of the relation between earthquake events, this solution was the one that worked the best for my purposes.

4 Tool overview

The GEA tool is a React Web application providing a dashboard that allows users to gain insights about the data by interacting with the system itself. The whole application has been coded by using several programming languages, such as JavaScript (and, in particular, the React.js framework and the D3.js library), and Python for the data acquisition and preprocessing.

4.1 Intended users

The target users are the types of people or organizations that would benefit from the insights provided by the dashboard. Here are some potential users and their objectives:

1. Seismologists and geophysicists researchers: these users can make use of the dashboard to study patterns of seismic activity across various regions, explore the correlation between depth, magnitude, and geographical distribution, and understand temporal patterns of earthquakes and volcanic eruptions or explosions. They might leverage the Parallel coordinates and t-SNE scatterplot to differentiate between event types and analyze specific conditions contributing to major events.
2. Government agencies and emergency response organizations: agencies responsible for earthquake preparedness and response can use the dashboard to identify high-risk areas based on past data understanding when and where high-magnitude events are more likely to occur. By tracking seismic activity over time, they can plan better resource allocation, such as where to position rescue teams, medical supplies, and equipment for rapid deployment.

3. Urban planners and civil engineers: these users can rely on the dashboard to identify stable regions with low seismic activity for building infrastructure like roads, bridges, and skyscrapers. They can also use insights on regional seismic risk to recommend seismic-resistant building codes and best practices for specific areas.

4.2 Employed visualizations

All the presented charts are interactive and each one offers the user the possibility to interact with the displayed data. The result of each interaction is reflected in all the other charts. For instance, if the user selects a cluster displayed on the scatterplot, the selected data will be shown in all charts. The Web application consists of a dashboard showing four charts: a geographical map, a scatterplot chart, a heatmap chart, and a parallel coordinates chart. A screen of the dashboard is reported in Figure 1.

Geographical map One of the primary features of GEA is its geographical map which displays seismic events on a global map. Each event is represented as a point, with size indicating magnitude category and color corresponding to depth category, allowing users to easily identify patterns such as clusters of shallow earthquakes or deep-seated seismic activity. The interactive map enables users to zoom in on specific regions or filter events by magnitude and depth, providing a detailed view of earthquake activity across tectonic plates and fault lines. This feature is useful for identifying seismic hotspots and understanding the spatial distribution of seismic events. The user can also select a specific portion of the map with a brushing action to highlight points in the map and analyze their features with the other charts. The chart is represented in the top left corner of Figure 1.

Parallel coordinates Another key component of GEA is its parallel coordinates plot, which facilitates the exploration of relationships between various seismic event attributes, such as event type, magnitude scale, reporting network, and distance to the nearest seismic station. This multi-dimensional view enables users to identify patterns and correlations that may not be immediately apparent in traditional single-variable analyses. For example, users can brush across axes to focus on specific ranges—such as examining how events with different magnitudes relate to reporting networks or studying the proximity of events to populated areas. The parallel coordinates plot is useful for identifying complex patterns in seismic data, which can inform targeted research, risk assessments, and disaster response planning. The chart is represented in the bottom left corner of Figure 1.

Heatmap Temporal analysis of seismic activity is important. GEA addresses this need through its heatmap, which displays seismic events over time, with weeks on the y-axis and months on the x-axis. Each cell in the heatmap represents the number of seismic events in a given week of the given month, offering a clear overview of how seismic activity fluctuates throughout the year using a linear graded legend that maps the number of events to a specific color gradient. By interacting with the heatmap, users can filter events by periods, allowing them to focus on high-activity weeks or months and correlate these periods with other variables, such as tectonic shifts or volcanic activity. This temporal analysis helps in identifying patterns such as seasonal fluctuations in seismic activity or surges in earthquake events following significant geological disturbances. The chart is represented in the top right corner of Figure 1.

Scatterplot To further enhance the exploration of seismic data, GEA includes a t-SNE scatterplot that provides a dimensionality-reduced representation of seismic events for clustering analysis. In this scatterplot, earthquakes are grouped with points color-coded according to magnitude categories. By this, users can easily differentiate between seismic events of different magnitude categories, providing a clear picture of how these characteristics are distributed across global seismic activity. This advanced visualization technique allows users to uncover clusters of events with similar properties, aiding in the classification of seismic events and providing insights into the underlying causes of seismic activity. The chart is represented in the bottom right corner of Figure 1.

5 Insights

The GEA dashboard provides users with the opportunity to gain valuable insights by delving into earthquake data through interactive visualizations. With the aid of four distinct charts, users can delve into the spatial, temporal, and multi-dimensional aspects of seismic events. This comprehensive approach enables the identification of intricate relationships between various key earthquake parameters. Furthermore, through the application of targeted filtering and brushing techniques, users can reveal significant patterns in seismic activity. These insights are invaluable for fostering a deeper understanding of global earthquake behavior and facilitating the formulation of informed data-driven decisions. Let's now proceed to observe the visualization tool in action to uncover detailed insights about the data.

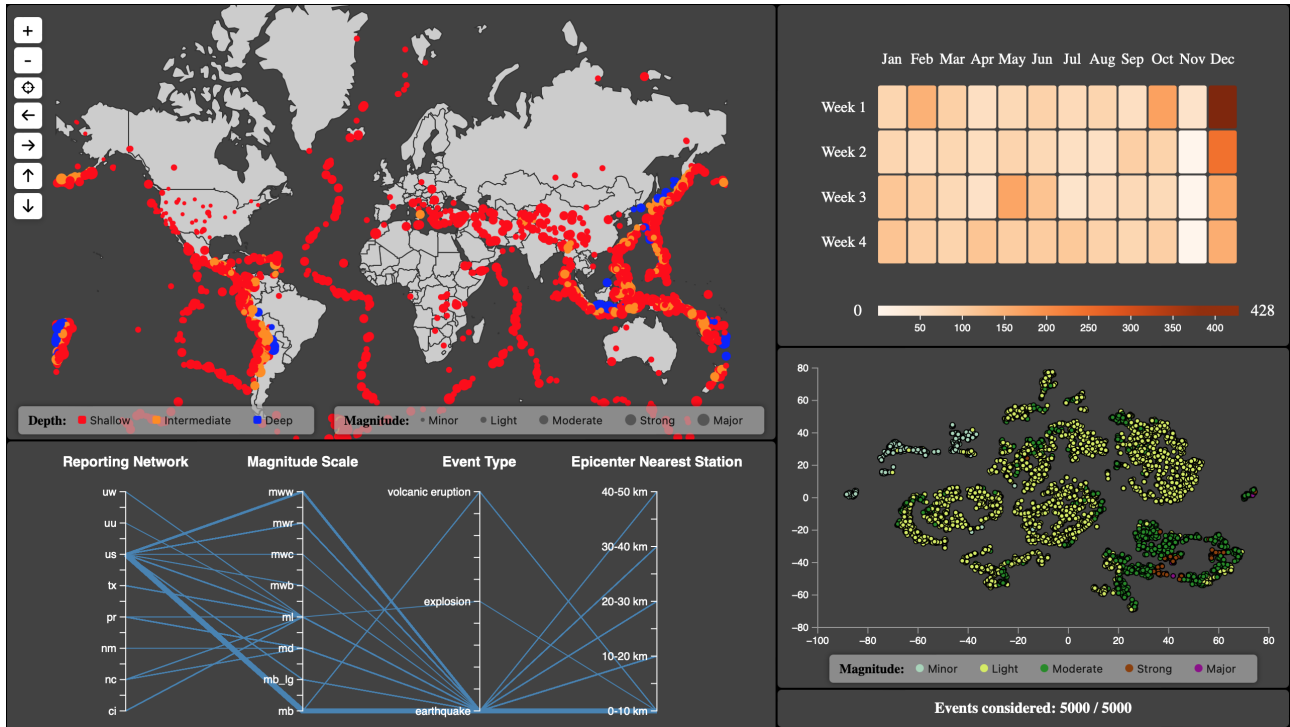


Figure 1: GEA dashboard

5.1 Regional seismic analysis

Figure 2 shows the result obtained by the user with the following workflow:

1. Geographical map: start by brushing a region on the map with the most seismic activity, focusing on this hotspot, like the one in the Pacific Ocean (i.e. eastern part of the map). This selection considers almost half of the available events.
2. Parallel coordinates: by filtering further on the epicenter's proximity to the nearest station, it's possible to down events to those occurring closer to seismic stations (i.e. 0-20 km).
3. Heatmap: selecting the month of December, a month with high seismic activity, it's possible to narrow the time frame of the analysis further.
4. t-SNE scatterplot: in this chart, it's clear how the events selected in the previous steps were highly correlated with light-magnitude earthquakes while there is also a small number of events correlated with minor-magnitude.

Focusing on areas with a high frequency of events on the map, highlighting regions that experience more seismic activity, could suggest a tectonically active zone. Being closer to seismic stations often results in more reliable measurements, lending credibility to the observed patterns. The correlation of these events with the light magnitude category implies that the region mostly experiences less severe, low-intensity earthquakes. These types of seismic events often go unnoticed by the general public or are viewed as less

dangerous. However, they can still play a significant role in disaster risk reduction strategies.

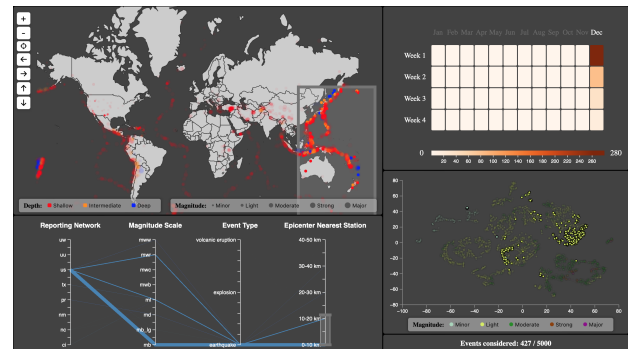


Figure 2: Regional seismic hotspot analysis

5.2 Deeper seismic events analysis

Figure 3 shows the result obtained by the user with the following workflow:

1. Geographical map: start by filtering events with a deep depth. It immediately becomes clear how these events are focused in the so-called "Pacific Ring of Fire".
2. Heatmap: these type of events rarely happens with a maximum of 13 events in a single week and December has the greater number of deeper events. However, week 4 is the one that contains an equal distribution of events during all months (i.e. a mean of 6 events per month), excluding November. For this reason, it is selected.

During the examination of seismic events, it's evident that deep earthquakes occur less frequently compared to shallow and intermediate ones. This is consistent with scientific understanding, as deep earthquakes are typically associated with tectonic activity in subduction zones, where one tectonic plate is forced deep into the Earth's mantle. Through parallel coordinates, it's possible to observe the correlation of these deep earthquakes with the Mw (Moment or Moment W-phase magnitude, calculated from the seismic moment) and Mb (Period body wave magnitude, based on P-waves) magnitude types, which are generally used for larger earthquakes. It's noteworthy that these events are exclusively reported by the United States network, due to its global network coverage and technological advancements obtained thanks to a historical investment in seismic research. Upon examining the t-SNE scatterplot, it's evident that these deep earthquakes are associated with medium to larger magnitudes, even though they are outliers in the chart. This suggests that when deep earthquakes do occur, they tend to release a significant amount of energy, resulting in higher-magnitude events. The depth of these earthquakes means that the seismic waves have to travel further to the surface, resulting in less intense shaking compared to shallow earthquakes of the same magnitude. This suggests that further research and monitoring in subduction zones may be valuable for understanding and preparing for deep earthquake events.

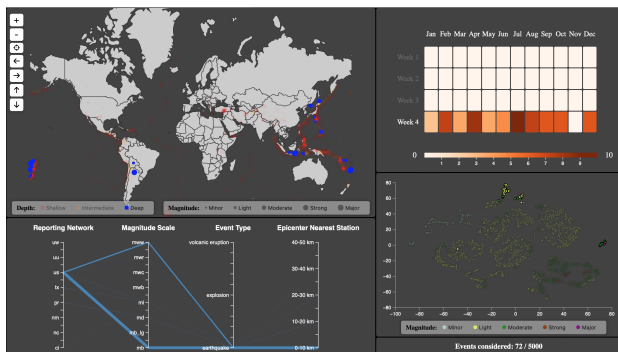


Figure 3: Deep seismic events analysis

5.3 Seismic hazard periodicity and regional impact analysis

Figure 4 shows the result obtained by the user with the following workflow:

1. t-SNE scatterplot: start by selecting the events with moderate, strong, and major magnitude categories. These magnitudes (ranging from 4.0 and above) are more likely to cause significant structural damage, making them important for this analysis.

2. Parallel coordinates: by filtering further on the epicenter's proximity to the nearest station, it's possible to down events to those occurring closer to seismic stations (i.e. 0-10 km). This helps concentrate the analysis on events that are not only well-recorded due to proximity to stations but also likely to affect nearby populated areas making the analysis more precise.
3. Geographical map: events that occur at shallow depths, defined as less than 70 kilometers, can have a more significant impact on the surface and pose a greater danger, particularly in densely populated areas. The intense shaking associated with these shallow earthquakes can compromise infrastructure, leading to potentially severe consequences. For this reason, the shallowest events are selected to focus the analysis only on them.

The heatmap chart demonstrates that significant earthquakes are evenly distributed across months from January to October, with no specific week standing out as a period of high activity. October has the highest number of significant earthquakes, making it a potentially more hazardous period. Conversely, August and November are quieter, with November showing activity only in the first week, suggesting that risk levels drop during these months. Based on these results, it can be concluded that the safer month is December, along with the last weeks of November, while the area related to the Pacific Ocean on both coasts is the most dangerous. This insight can guide emergency response preparations and urban planning, helping to prioritize resources during higher-risk periods. It can also be useful to explore if these results depend on other elements, like tectonic, seasonal, and environmental/atmospheric factors.

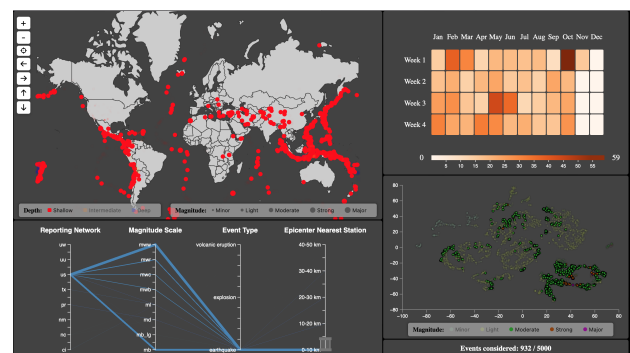


Figure 4: Seismic hazard periodicity and regional impact analysis

6 Conclusions

This paper focused on the presentation of GEA (Global Earthquakes Analytics), a comprehensive and interactive visualization tool designed to enhance understanding of global seismic activity. GEA utilizes various data visualization techniques, including a geographical map, parallel coordinates, heatmap, and t-SNE scatterplot, to allow users to explore earthquake data based on magnitude, depth, geographical distribution, and time. The tool provides an easy-to-use interface with powerful filtering and brushing capabilities for detailed exploration of seismic patterns and correlations. Unlike existing tools, GEA offers a multi-dimensional approach that supports complex analysis, enabling users to uncover intricate relationships between seismic characteristics. This enhances earthquake risk assessment and aids in disaster preparedness efforts. By integrating advanced analytical techniques, such as dimensionality reduction, with user-friendly interactive features, GEA is suitable for scientific research and data-driven decision-making for earthquake mitigation. The tool not only supports the study of seismic patterns and trends but also contributes to improving public safety and resilience in earthquake-prone regions. In the future, GEA could be expanded to include real-time data streams or integrated with predictive models, further enhancing its utility for researchers and emergency management professionals.

References

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